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Calc environment ready.
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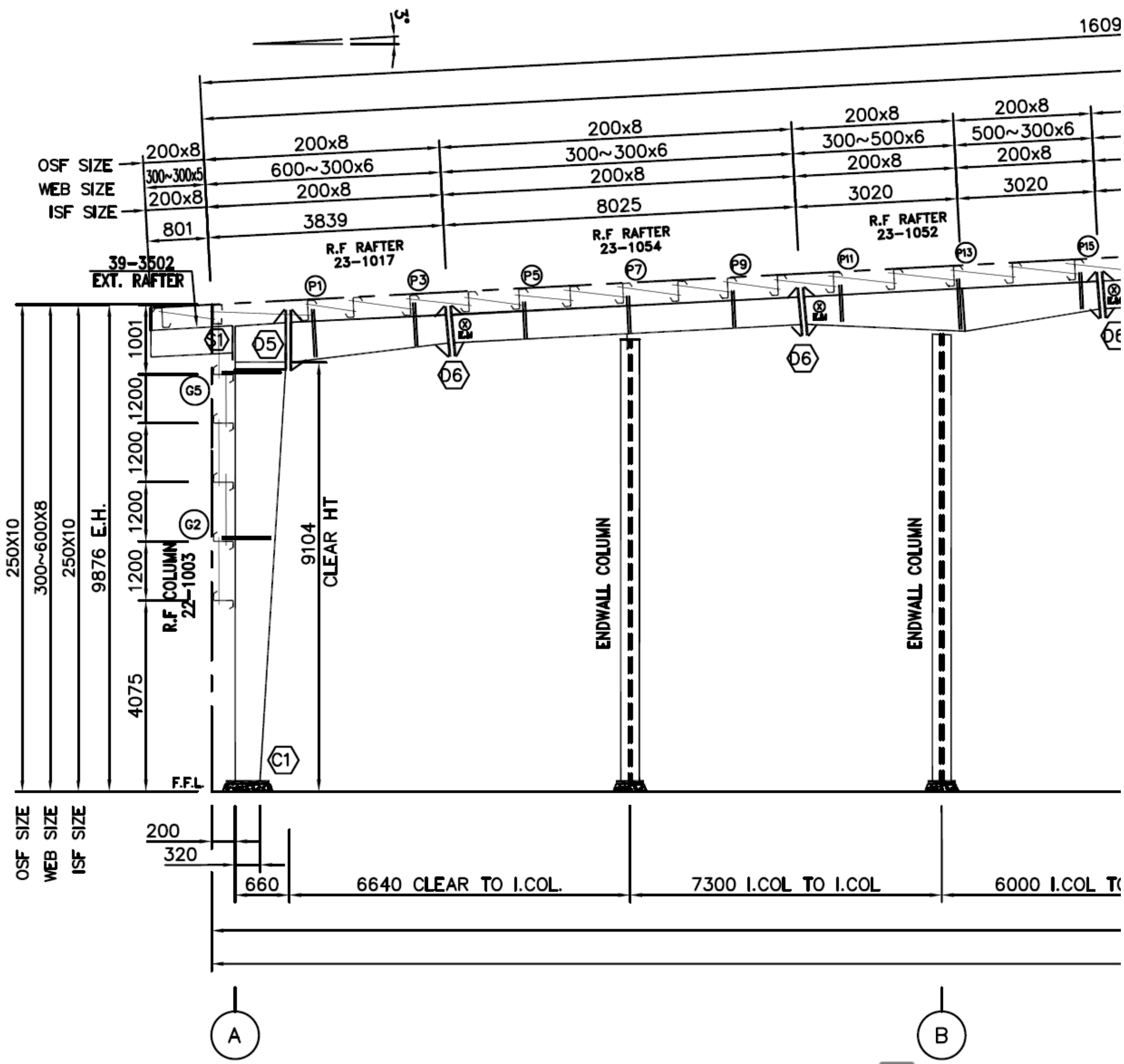
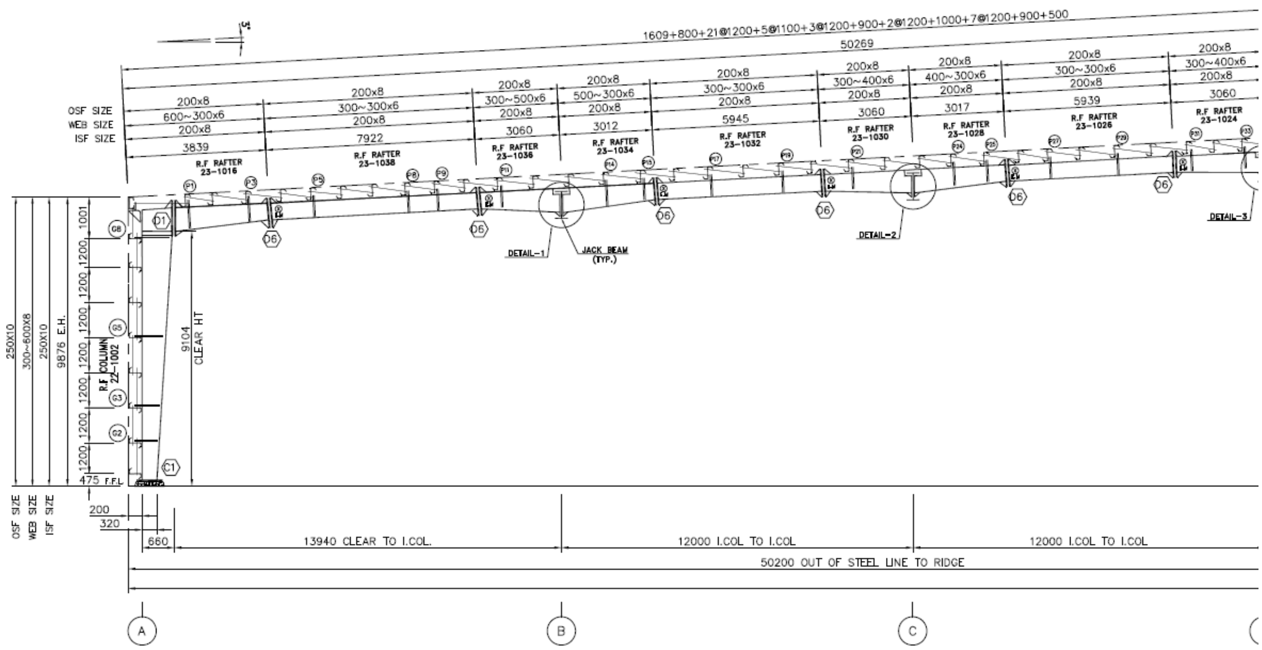
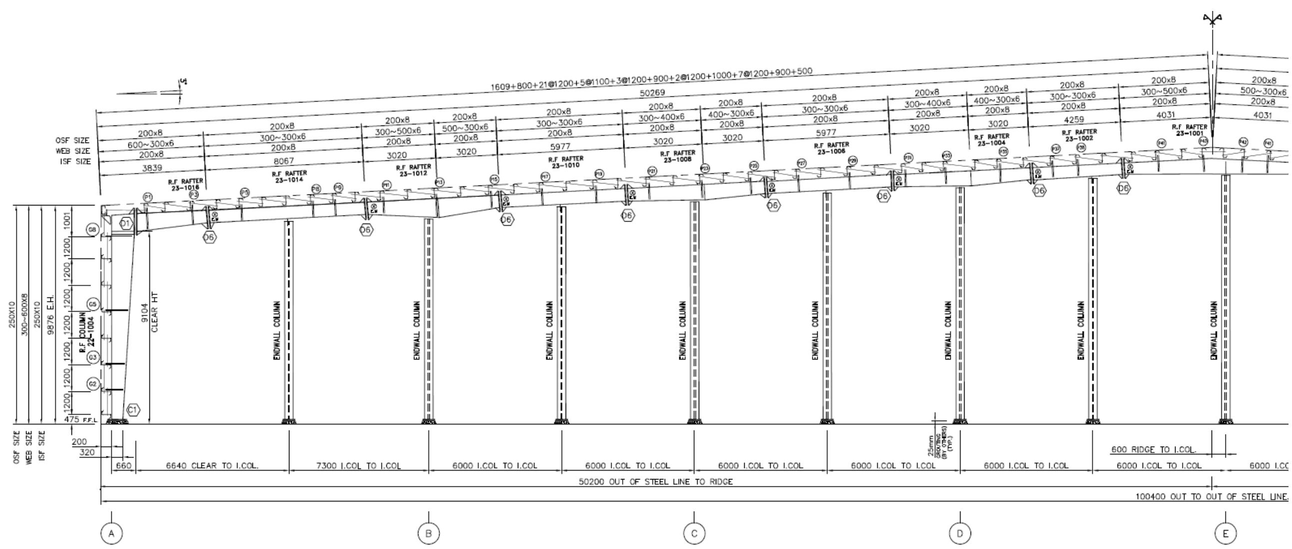
Member Capacity Calculation

Purlin design check







CROSS SECTION
(@ GL.11)

* NOTE :

- DETAILS D1, D5, D6, S1 REFER ON XXX-1003-F.DWG

$\text{fy} = 450$ (steel yield strength (MPa) typical steel grade is G550. using G450 for more conservative design)

$t = 2$ (thickness (mm))

$$Z = Z_{cm3} \cdot 1 \times 10^{3} = 32 \cdot 1 \times 10^{3} = 32000.000 \quad \text{(convert cm3} \rightarrow \text{mm3)}$$

$$M_{Rd} = \frac{\text{fy} \cdot Z}{\gamma_{M0} \cdot 1 \times 10^{6}} = \frac{450 \cdot 32000.000}{1.000 \cdot 1 \times 10^{6}} = 14.400$$

$$A_v = t \cdot h = 2 \cdot 200 = 400$$

$$V_{Rd} = \frac{A_v \cdot \text{fy}}{\text{math.sqrt3} \cdot \gamma_{M0}} \cdot \frac{1}{1000} = \frac{400 \cdot 450}{math.sqrt3 \cdot 1.000} \cdot \frac{1}{1000} = 103.923$$

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simply supported with 1 concentrated loads

$L = 8.325$ (span (m)) $\quad P = 1.500$ (each point load (kN)) $\quad P_{ult} = 2.250$ (factored load (kN))

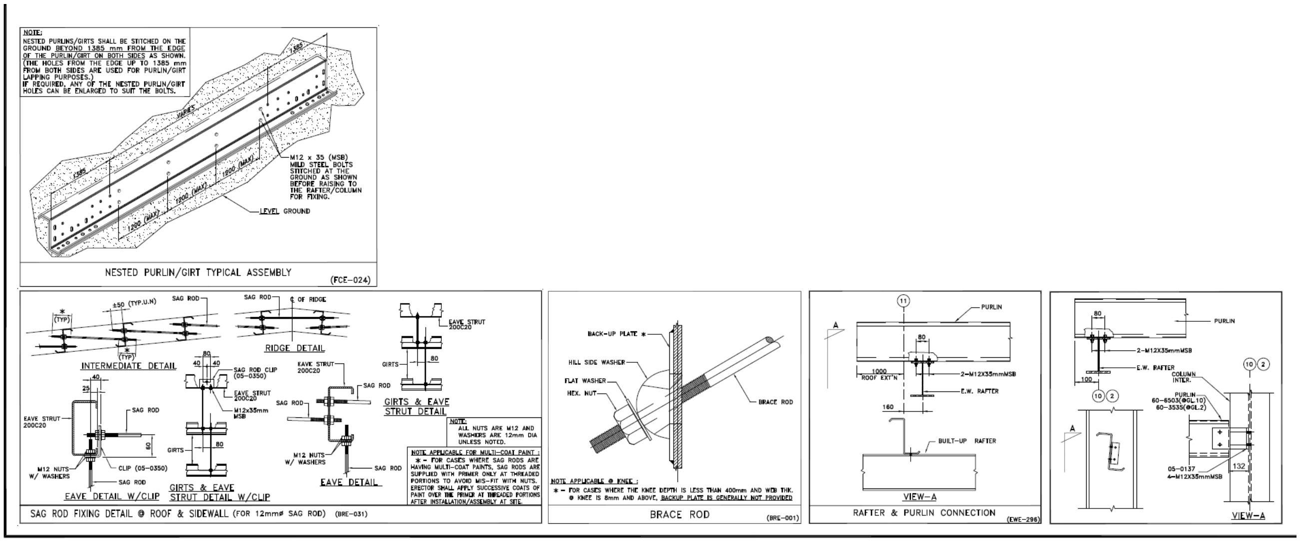
$$R = \frac{P_{ult}}{2} = \frac{2.250}{2} = 1.125$$

$$M_{max} = P_{ult} \cdot \frac{L}{4} = 2.250 \cdot \frac{8.325}{4} = 4.683$$

$$\text{UR}_{bending} = \frac{M_{max}}{M_{Rd}} = \frac{4.683}{14.400} = 0.325 \quad \text{(<1 hence bending Pass)}$$

$$\text{UR}_{shear} = \frac{R}{V_{Rd}} = \frac{1.125}{103.923} = 0.011 \quad \text{(<1 hence shear Pass)}$$

Connection design check



$L = 6.000$ (purlin span (m)"”) $s = 0.600$ (purlin spacing (m)) $q = ($

$\text{wind}_{uplift} = 1.200$ (wind suction (kN/m2)) $d = 12$ (bolt diameter (mm)) $A_b = 8$

$f_{ub} = 400$ (bolt ultimate strength (MPa)) $\gamma_{M2} = 1.250$ (EC3 bolt partial factor) $\alpha_v = ($

$t = 2$ (purlin thickness (mm)) $f_u = 550$ (ultimate steel strength (MPa)) $n_{bolts} = 2$

$$V_{b_{Rd}} = \frac{\alpha_v \cdot f_{ub} \cdot A_b}{\gamma_{M2} \cdot 1000} = \frac{0.600 \cdot 400 \cdot 84.300}{1.250 \cdot 1000} = 16.186$$

$$V_{connection_{Rd}} = n_{bolts} \cdot V_{b_{Rd}} = 2 \cdot 16.186 = 32.371$$

$$N_{b_{Rd}} = \frac{0.9 \cdot f_{ub} \cdot A_b}{\gamma_{M2} \cdot 1000} = \frac{0.9 \cdot 400 \cdot 84.300}{1.250 \cdot 1000} = 24.278$$

$$N_{connection_{Rd}} = n_{bolts} \cdot N_{b_{Rd}} = 2 \cdot 24.278 = 48.557$$

$$F_{b_{Rd}} = \frac{2.5 \cdot d \cdot t \cdot f_u}{\gamma_{M2} \cdot 1000} = \frac{2.5 \cdot 12 \cdot 2 \cdot 550}{1.250 \cdot 1000} = 26.400$$

$$F_{b_{connection_{Rd}}} = n_{bolts} \cdot F_{b_{Rd}} = 2 \cdot 26.400 = 52.800$$

$$\text{UR}_{shear} = \frac{R}{V_{connection_{Rd}}} = \frac{1.125}{32.371} = 0.035 \text{ (connection Pass, shear failure mode)}$$

$$\text{UR}_{tension} = \frac{R}{N_{connection_{Rd}}} = \frac{1.125}{48.557} = 0.023 \text{ (connection Pass, tension failure mode)}$$

$$\text{UR}_{bearing} = \frac{R}{F_{b_{connection_{Rd}}}} = \frac{1.125}{52.800} = 0.021 \text{ (connection Pass, bearing failure mode)}$$